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Semiconductor Capital Equipment | North America

Thoughts on Intel Capex

There's upside to our \$25bn assumption for 2027 WFE. We see TEL, Lasertec and KLA as core beneficiaries.

Key Takeaways

- Our \$202bn (+31% y/y) 2027 WFE forecast reflects \$25bn of Intel capex.
- 22.4mn units of Xeon6/7 would require \$24-28bn of WFE.

Inevitability to Intel Capex revisions. Intel revised up their 2026 capex outlook to \$20bn+ and spoke of 2027 capex to be "significantly above 2026 levels". As highlighted in [Semiconductor Capital Equipment: 2Q'26 WFE update, memory mismatch unresolved](#) (18 May 2026), our view was that were Intel to take part in this CPU upcycle, we think there is an inevitability to capex hikes. We estimated that if the company wants to produce 5mn incremental Xeon 6s, the company will have to spend \$5.3bn in WFE. Assuming Intel wants to take two-thirds of the incremental CPU units, we think the company will have to expand logic wafer capacity by a minimum of \$23.8bn. Assuming 2 compute tiles for Xeon 7, we estimate Intel will have to spend \$27.9bn in WFE. We saw capex tracking possibly even higher as our capex argument is purely based on Intel looking to increase capex to meet rising CPU demand. Intel IDM. Foundry customers acquisition will necessitate spending beyond the \$23.8-27.9bn of WFE to meet rising CPU demand. Our [latest WFE forecast](#) reflects \$25bn of Intel capex in 2027, but there's upside.

Biggest beneficiaries outside of the US. We see the biggest beneficiaries to an Intel capex hike outside of the US, as Tokyo Electron has seen Intel as a 10%+ customer from F3/14-24, reaching as high as 20.4% in F3/20. Similarly, Lasertec has seen Intel as a 10% customer from F6/20-25, reaching as high as 31.6% in F6/22. **Within the US we see KLA as an outsized beneficiary.** Intel has not been a 10% customer for KLA since F6/15, but we think Intel is leaning on KLA as a key partner to improve yield in order to improve margin and become a viable foundry. We see possibility of KLA 6.4x their Intel revenue from 2025 to 2027 on the back of PC intensity and share gain. In our view, this update from Intel validates our positive view on KLA's long-term story, but as we highlighted in our [preview](#), we would be slightly more cautious than positive into the print. *Tokyo Electron and Lasertec are covered by Suzune Tamura.*

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Wafers per CPU

Exhibit 1: We estimate 5mn incremental server CPUs would require \$5.3bn+ of WFE

WFE per CPU Needs (\$mn)	Intel Xeon 6	Intel Xeon 7
Total	5,326	6,233
Logic	5,251	6,158
2nm	0	4,535
3nm	4,502	0
7nm	749	1,623
EMIB	75	75

Wafers per CPU Needs		
Total	23,487	24,214
Logic	22,485	23,213
2nm	0	14,199
3nm	18,325	0
7nm	4,160	9,014
EMIB	1,001	1,001

Source: Morgan Stanley Research estimates. 2nm = Intel 18A, 3nm = Intel 3.

Exhibit 2: Two-thirds of the incremental units through 2030 (22.4mn units) would require \$23.8bn+ of WFE

WFE per CPU Needs (\$mn)	Intel Xeon 6	Intel Xeon 7
Total	23,854	27,913
Logic	23,517	27,577
2nm	0	20,311
3nm	20,164	0
7nm	3,354	7,266
EMIB	336	336

Wafers per CPU Needs		
Total	105,182	108,441
Logic	100,698	103,957
2nm	0	63,588
3nm	82,066	0
7nm	18,632	40,368
EMIB	4,484	4,484

Source: Morgan Stanley Research estimates. 2nm = Intel 18A, 3nm = Intel 3.

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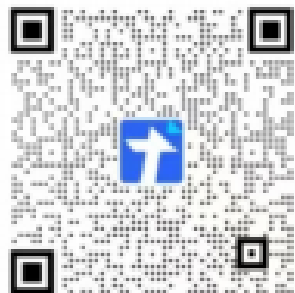
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Valuation Methodology and Risks

Lasertec (6920.T)

P/E is set in line with front-end equipment peers under our coverage.

Risks to Upside

- Strong uptake of the new product
- Rising demand for EUV mask inspection systems driven by robust investment in leading-edge semiconductors.
- Higher ASPs and replacement demand driven by the transition to HNA systems.

Risks to Downside

- Suspension of advanced semiconductor development by major semiconductor manufacturers and increased competition from new market entrants
- Weak performance of the new product launch

Tokyo Electron (8035.T)

Our target P/E uses the average one-year forward weekly multiple (just under 30x) since October 2025, when the current SPE upcycle began, through to the present.

Risks to Upside

- Intel & TeraFab capex materializes sooner than expected
- Expansion in applications of NAND for AI inference, and resulting acceleration in green-field investments
- Earlier-than-expected ramp in mass production adoption of cryo etchers.

Risks to Downside

- Export controls to China tighten
- Coater/developer market shrinks due to adoption of dry resist technology
- Tech innovation greatly reduces memory usage in AI inference

Intel Corporation (INTC.O)

~42x EPS CY2027 EPS of \$1.77, 42x is above the high end of the large cap logic semi peer group, reflecting high leverage potential on numbers that are still depressed, and foundry optionality, despite our longer term skepticism

Risks to Upside

- Foundry partnerships de-risk the story and further improve the multiple
- The company regains lost share in desktop and server following CPU shortages

Risks to Downside

- AMD competition increasingly becomes more significant, which could lead to further share losses in processors and pressure on ASPs
- Minimal success in foundry leads to an inflated cost structure

KLA Corp (KLAC.O)

38x CY28e EPS of \$7.22, a 15% premium to US SPE peers LAM & AMAT to reflect growth prospects in foundry logic and DRAM.

Risks to Upside

- Foundries (TSMC, Intel, Samsung) pull forward their logic roadmaps
- Design starts in leading edge nodes increase
- Process control intensity increases in DRAM due to HBM

Risks to Downside

- Certain foundries (Intel, Samsung) scale back their aspirations
- Market share loss in mask inspection to Lasertec or in e-beam to AMAT or ASML
- China restrictions

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(as of June 30, 2026)

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Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1544	42%	453	49%	29%	757	44%
Equal-weight/Hold	1577	43%	390	42%	25%	769	44%
Not-Rated/Hold	3	0%	1	0%	33%	1	0%
Underweight/Sell	544	15%	89	10%	16%	204	12%
Total	3,668		933			1731	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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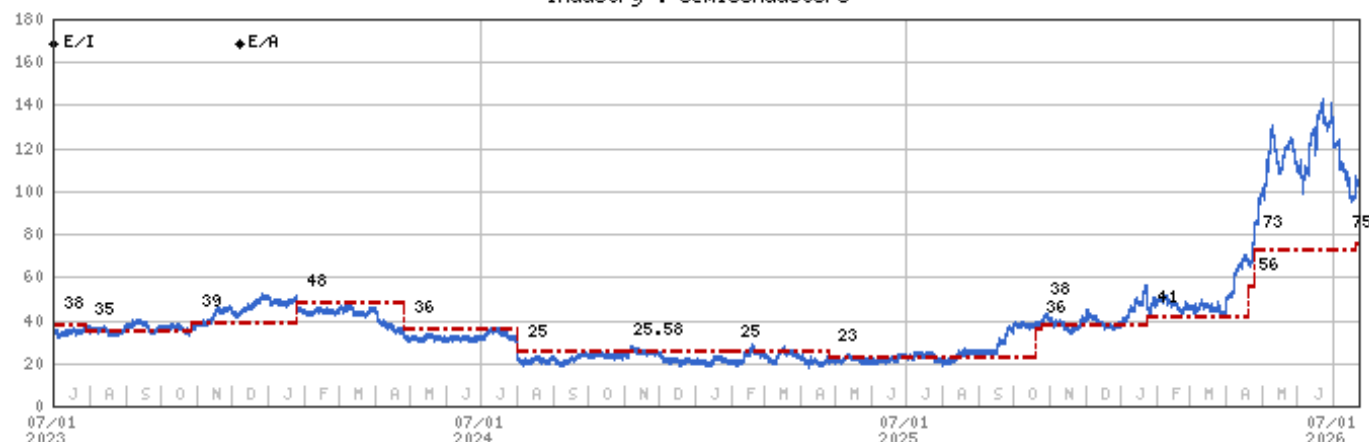
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Stock Price, Price Target and Rating History (See Rating Definitions)

Intel Corporation (INTC.O) - As of 07/23/26 GMT in USD
Industry : Semiconductors



Stock Rating History: 7/1/21 : O/I; 10/22/21 : E/I; 3/3/22 : U/I; 2/22/23 : E/I; 12/1/23 : E/A

Price Target History: 1/22/21 : 72; 7/23/21 : 70; 10/5/21 : 67; 10/22/21 : 55; 3/3/22 : 47; 4/28/22 : 46; 7/28/22 : 36; 10/27/22 : 29.5; 2/22/23 : 28; 4/27/23 : 31; 6/15/23 : 38; 7/28/23 : 35; 10/27/23 : 39; 1/26/24 : 48; 4/26/24 : 36; 8/2/24 : 25; 10/31/24 : 25.58; 1/31/25 : 25; 4/25/25 : 23; 10/20/25 : 36; 10/24/25 : 38; 1/23/26 : 41; 4/20/26 : 56; 4/24/26 : 73; 7/20/26 : 75

Source: Morgan Stanley Research Date Format: MM/DD/YY Price Target --- No Price Target Assigned (NA)

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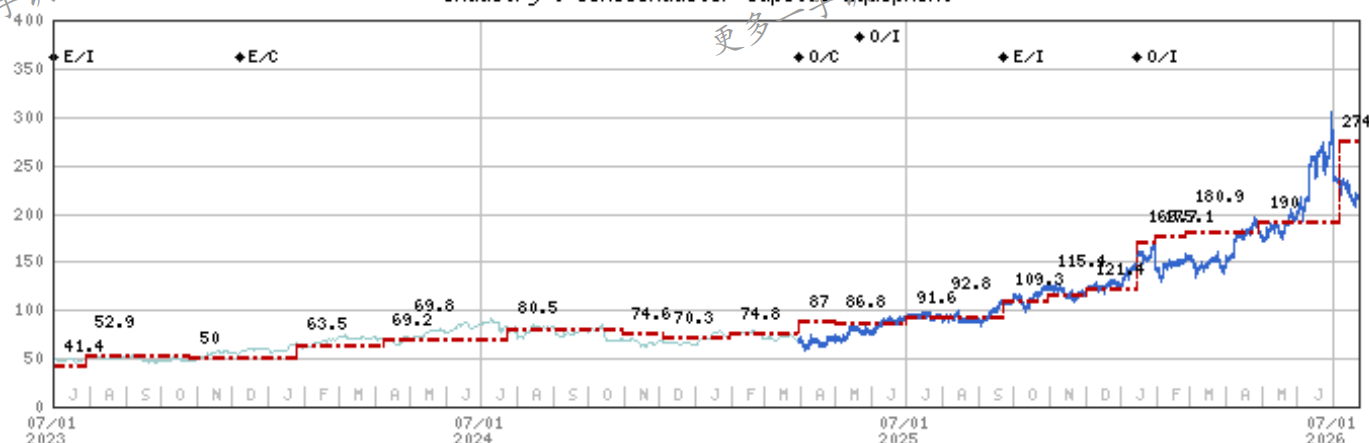
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Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

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KLA Corp (KLAC.O) - As of 07/23/26 GMT in USD
Industry : Semiconductor Capital Equipment



Stock Rating History: 7/1/21 : E/I; 12/1/23 : E/C; 3/30/25 : O/C; 5/22/25 : O/I; 9/22/25 : E/I; 1/15/26 : O/I

Price Target History: 4/30/21 : 36.4; 7/30/21 : 38.9; 10/28/21 : 39.9; 12/14/21 : 43.1; 1/28/22 : 40; 4/25/22 : 37.5; 4/29/22 : 37.3; 7/24/22 : 37.7; 7/28/22 : 38.1; 9/6/22 : 35.8; 10/17/22 : 28.9; 10/27/22 : 31.8; 1/27/23 : 42; 4/26/23 : 41.4; 7/28/23 : 52.9; 10/26/23 : 50; 1/26/24 : 63.5; 4/8/24 : 69.2; 4/26/24 : 69.8; 7/24/24 : 80.5; 10/30/24 : 74.6; 12/5/24 : 70.3; 1/31/25 : 74.8; 3/30/25 : 87; 5/1/25 : 86.8; 7/1/25 : 91.6; 7/31/25 : 92.8; 9/22/25 : 109.3; 10/29/25 : 115.4; 12/1/25 : 121.4; 1/15/26 : 169.7; 1/29/26 : 175.1; 2/24/26 : 180.9; 4/29/26 : 190; 7/6/26 : 274

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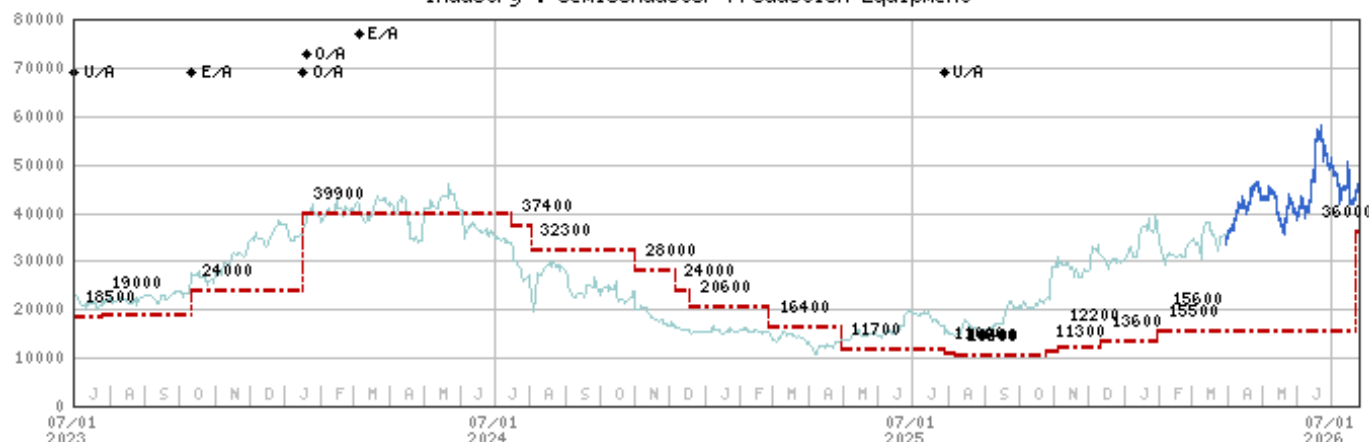
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Lasertec (6920.T) - As of 07/24/26 GMT in JPY
Industry : Semiconductor Production Equipment



Stock Rating History: 6/19/23 : U/A; 10/11/23 : E/A; 1/17/24 : O/A; 1/19/24 : O/A; 3/5/24 : E/A; 7/28/25 : U/A

Price Target History: 6/19/23 : 18500; 7/25/23 : 19000; 10/11/23 : 24000; 1/17/24 : 39900; 7/16/24 : 37400; 8/2/24 : 32300; 11/1/24 : 28000; 12/5/24 : 24000; 12/19/24 : 20600; 2/25/25 : 16400; 4/29/25 : 11700; 7/28/25 : 11000; 8/7/25 : 10500; 8/8/25 : 10400; 10/24/25 : 11300; 11/4/25 : 12200; 12/12/25 : 13600; 1/30/26 : 15500; 2/2/26 : 15600; 7/22/26 : 36000

Source: Morgan Stanley Research Date Format: MM/DD/YY Price Target --- No Price Target Assigned (NA)

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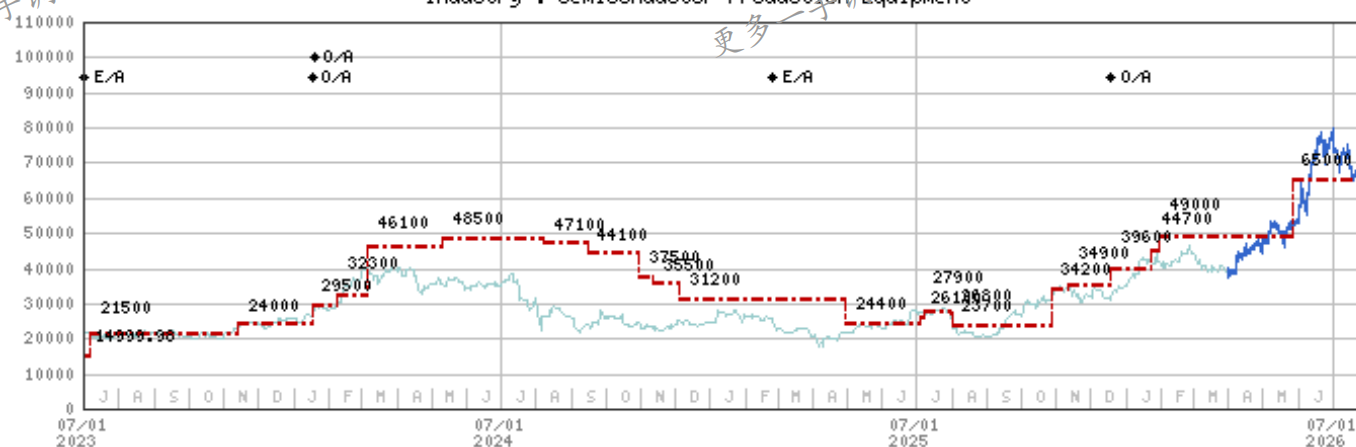
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Tokyo Electron (8035.T) - As of 07/24/26 GMT in JPY
Industry : Semiconductor Production Equipment



Stock Rating History: 7/1/21 : O/A; 7/26/22 : E/A; 1/17/24 : O/A; 1/19/24 : O/A; 2/25/25 : E/A; 12/19/25 : O/A

Price Target History: 4/19/21 : 17999.98; 10/25/21 : 18999.98; 11/15/21 : 20999.98; 6/8/22 : 21333.31; 7/26/22 : 15999.98; 10/18/22 : 11666.65; 2/10/23 : 14999.98; 7/7/23 : 21500; 11/13/23 : 24000; 1/17/24 : 29500; 2/9/24 : 32300; 3/6/24 : 46100; 5/11/24 : 48500; 8/8/24 : 47100; 9/15/24 : 44100; 10/30/24 : 37500; 11/12/24 : 35500; 12/5/24 : 31200; 4/30/25 : 24400; 7/4/25 : 26100; 7/7/25 : 27900; 7/31/25 : 26800; 8/1/25 : 23700; 10/27/25 : 34200; 11/11/25 : 34900; 12/19/25 : 39600; 1/23/26 : 44700; 1/30/26 : 49000; 5/27/26 : 65000

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INDUSTRY COVERAGE: Semiconductor Capital Equipment

COMPANY (TICKER)	RATING (AS OF)	PRICE* (07/23/2026)
Shane Brett		
Advanced Energy Industries Inc. (AEIS.O)	O (07/22/2026)	\$315.14
Applied Materials Inc. (AMAT.O)	E (05/18/2026)	\$562.80
Camtek (CAMT.O)	E (12/01/2025)	\$155.19

KLA Corp (KLAC.O)	O (01/15/2026)	\$218.73
Lam Research Corp (LRCX.O)	O (05/18/2026)	\$319.78
MKS Inc. (MKSI.O)	O (08/04/2024)	\$345.30
Nova Ltd (NVMI.O)	E (12/01/2025)	\$455.37
ONTO Innovation Inc (ONTO.N)	O (06/14/2026)	\$291.27
Teradyne Inc (TER.O)	E (07/30/2025)	\$373.75

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